

ROC Time Slice Bank

Header

Stream Info Bank (SIB)

Data Banks

EVIO HEADER		
ROC Bank Length		
ROC ID	0x10	SS
Stream Info Length		
0xFF30	0x20	SS
0x31	0x01	TSS Len
Frame Number		
Timestamp1 (31 – 0)		
Timestamp1 (63 – 32)		
0x41	0x85	AIS Len
Payload 2		Payload 1
...		...
0		Payload N
Payload Port 1 Length		
PP 1 ID	0xf	SS
PP 1 Data		
...		
Payload Port N Length		
PP N ID	0xf	SS
PP N Data		

16 bit ROC ID Bank of banks

ROC ID	0x10	SS	Stream Status
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Err	Total Streams	Stream Mask
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Bits 7 6 - 4 3 - 0

Time Slice Segment (TSS)

2nd byte: top 2 bits = 2 for padding if odd # payloads (else 0), lower 6 bits is type 5 (unsigned short) => 0x85 (1000 0101)

Aggregation Info Segment (AIS)

Err	ND	DD		Module ID	Bond	Lane ID	Payload Port #
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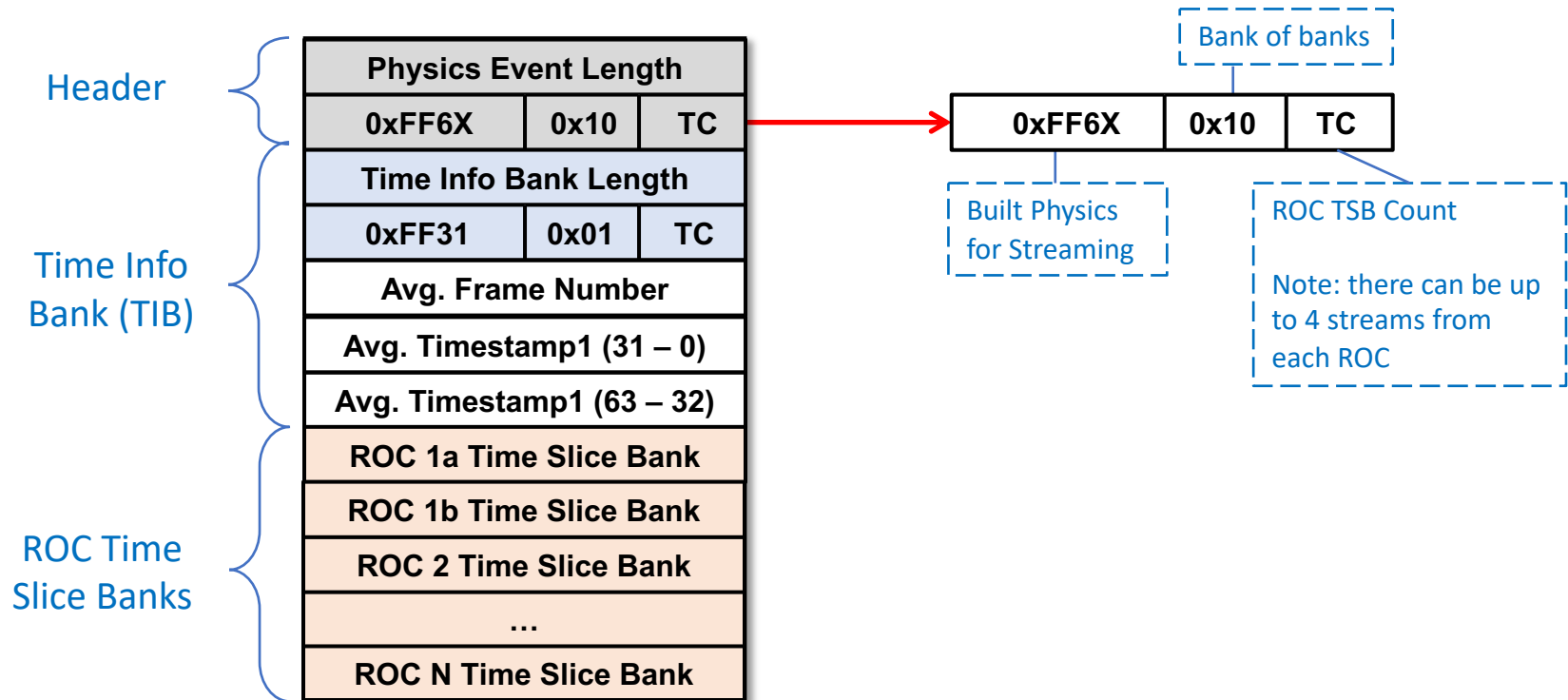
15 14 13 12 11 - 8 7 6, 5 4 - 0

ND = No Data Available DD = Data was Discarded

Payload Port (SS) :					7	6-3	2	1	0
Err	Mode			DD	ND				

Multiple Time Slices can be transported as defined in the EVIO Header. Multiple Streams can be included as defined by the Stream Status bits.

Streaming Physics Event



STREAMING TAGS

Tag Value	Purpose
0xFF30	Stream Info Bank (SIB) from ROC containing frame # followed by timestamp
0xFF31	Stream Info Bank (SIB) from Aggregator containing frame # followed by timestamp
0xFF32	Stream Info Bank (SIB) from Aggregator containing frame # followed by timestamp with non-fatal error
0x31	Time Slice Segment (TSS) from ROC containing frame # followed by timestamp
0x41	Aggregation Info Segment (AIS) from ROC
0x32	Time Slice Segment (TSS) from Aggregator containing frame # followed by timestamp
0x42	Aggregation Info Segment (AIS) from Aggregator
0xFF60	Built from ROC Raw Data streamed from ROC / VTP
0xFF61	Built from ROC Raw Data streamed from ROC / VTP with non-fatal error
0xFF62	Possibly: event built by DC event builder in streaming mode
0xFF64	Possibly: event built by secondary event builder in streaming mode
0xFF66	Possibly: event built by primary event builder in streaming mode
0xFF68	Missing Frame (no ROC data available, estimate for timestamp)

Event Building with slices

